

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Technology (M.Tech)

Semester – II University Examination June- 2010

M.Tech. (C.S.E.)

EC – 710 Principles of Wireless Networks

Date: 24.06.10, Thursday

Time: 12:30 p.m to 03:30 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Answer to the two sections must be written in the separate answer books.
3. Figures to the right indicate *full* marks.
4. Make suitable assumptions and draw the neat figures wherever required

SECTION- I

- Q1 (a) Distinguish between synchronous & asynchronous modes of communication by MAC protocols. 02
- (b) What are the possible challenges Adhoc Network can possibly face? Explain any two of them in detail. 06
- (c) Explain hidden terminal problem with possible solution of it. 04
- Q2 (a) In detail explain multiple access with collision avoidance with help of necessary diagram (MACA). 06
- (b) Conceptually Explain Merging & Fragmenting. 04

OR

- Q2 (a) Design Power aware logic utilized in Power-aware multi-access protocol with signal. 06
- (b) What is Adhoc Network? Explain Hetrogeneity in mobile devices. 04
- Q3 (a) With help of necessary figure illustrate hand shake mechanism utilized in MARCH (Media Access with reduced Hand Shake) protocol. 06
- (b) Explain the architecture and components of PRNET (Packet Radio network). 04
- (c) What is a *scope* in Fisheye State Routing? 03

OR

- Q3 (a) What is wireless sensor networks? Explain different traffic profile in wireless network. 06
- (b) Write short note on three frame packet forwarding in PRNET. 04
- (c) Write short note on Network Coding 03

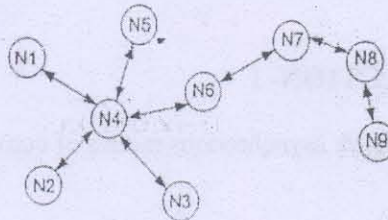
SECTION – II

Q4 (a) Give a possible classification of wireless multicast protocols. Explain any one of multicast routing protocol in detail. 07

(b) Describe Broadcast Incremental Power (BIP) protocol. 05

Q5(a) Look at the figure below. Construct the route table advertised by node N4 05

If DSDV is used as the routing protocol (three columns: Destination, Metric and Sequence Number).



(b) How the multicast mesh structure and the overlay structure can increase the robustness? 05

OR

Q5 (a) Give the list of stochastic mobility models. Explain Random Waypoint (RWP) Mobility Model in detail 07

(b) What is the difference between proactive routing and reactive routing? 03

Q6 (a) What are the major challenges for routing in MANET? Explain the operation of DSDV (Distance Vector Routing Protocol) in MANET. 08

(b) Discuss Landmark Ad Hoc Routing and Optimized Link state Routing in brief. 05

OR

Q6 (a) Which are the research scopes in congestion control in the area of heuristic Protocol development 06

(b) Write a short note on Minimum Spanning Tree Graph 05

(c) What is a landmark in LANMAR 02